Order Service Architecture

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1. Overview

The Order Service (order-service) is owned by the Orders team, with Mateo Rossi as technical lead. It is implemented in Java 17 with Spring Boot 3 and listens on port 8400. Service-level objectives are reviewed quarterly with product owners. Tracing uses OpenTelemetry with a ten percent head-based sampling rate. On-call handover happens every Monday at 10:00 local time.

Secrets must never be committed; the pre-commit hook scans for high-entropy strings. Structured JSON logging is mandatory; free-text logs are dropped by the log pipeline. Database schema migrations must be backward compatible for at least one release. Structured JSON logging is mandatory; free-text logs are dropped by the log pipeline.

2. Data Stores

Primary data store: PostgreSQL orders-db with a transactional outbox table. Infrastructure changes go through Terraform plans that need two approvals. Every service has a README that links to its dashboards and runbook.

3. Dependencies

Order Service depends on: payment-service, inventory-service, orders-db, Kafka topic orders.events. Container images are built once and promoted across environments by digest. Load tests should be coordinated in the platform channel so they do not trip alerts.

4. Security and Authentication

Authentication mechanism: Order Service uses mTLS between services plus OAuth2 client-credentials tokens. Tracing uses OpenTelemetry with a ten percent head-based sampling rate. Load tests should be coordinated in the platform channel so they do not trip alerts.

5. Scaling

Flaky integration tests in CI are tracked on the quality board; please do not just re-run them. Feature work is paused during incident follow-up weeks until action items are closed. Feature work is paused during incident follow-up weeks until action items are closed. Remember to tag pull requests with the owning team so reviews get routed correctly. Staging mirrors production topology but runs at roughly one tenth of the capacity.

We should revisit the alert thresholds after the next game day. Postmortems are blameless and are published within five business days. Database schema migrations must be backward compatible for at least one release. Feature work is paused during incident follow-up weeks until action items are closed.

6. Open Questions

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